

0.1 Preliminary

0.1.1 Direct Sum

Definition 0.1.1. Let V be a Vector Space, and $S_1, S_2 \subset V$ be subsets. Define *Sum* of S_1 and S_2 as:

$$S_1 + S_2 \stackrel{\text{def}}{=} \{a + b \mid a \in S_1, b \in S_2\}$$

In general, the sum of sets is not space, but sum of subspaces is the smallest subspace containing the spaces.

Definition 0.1.2. Let V be a vector space, and $W_1, \dots, W_n \leq V$ be subspaces.

We say that $\{W_1, \dots, W_n\}$ are *independent* if: for each $\alpha_i \in W_i$,

$$\alpha_1 + \dots + \alpha_n = 0 \implies \alpha_1 = \dots = \alpha_n = 0$$

Definition 0.1.3. Let V be a vector space.

If V is a sum of independent collection of subspaces $\{W_1, \dots, W_n\}$, then we say that V is *Direct sum* of the subspaces.

0.1.2 Projection

Definition 0.1.4. Let V be a vector space. A linear operator $E: V \rightarrow V$ is called a *projection* if $E^2 = E$.

Proposition 0.1.5. Let $E: V \rightarrow V$ be a projection on a vector space V . Then, the followings are satisfied:

1. For any $x \in V$, $x \in \text{im } E \iff E(x) = x$.
2. $V = \text{im } E \oplus \ker E$.
3. If V has finite dimension with β_1 is basis for $\text{im } E$ and β_2 is basis for $\ker E$, then

$$[E]_{\beta_1 \cup \beta_2} = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$$

where $r = \text{rank } E$.

Proof. Let $x \in \text{im } E$ be given. Then, $x = E(y)$ for some $y \in V$. Now,

$$E(x) = E(E(y)) = E(y) = x$$

And the converse is trivial. The second assertion is shown by: Given $x \in V$,

$$x = E(x) + (x - E(x))$$

Clearly $E(x) \in \text{im } E$ and $E(x - E(x)) = E(x) - E^2(x) = E(x) - E(x) = 0$, thus $x - E(x) \in \ker E$. Further, since $E(0) = 0$, the independence is clear. □

Theorem 0.1.6. Let V be a vector space over F .

If $V = W_1 \oplus \cdots \oplus W_k$, then there exists k linear operators E_i on V such that

$$E_i \circ E_j = \delta_{ij} E_i, \quad I = E_1 + \cdots + E_k, \quad W_i = \text{im } E_i$$

for each $1 \leq i \leq k$.

Proof. Suppose that V is a direct sum of $W_1, \dots, W_k$. For each $1 \leq j \leq k$, define

$$E_j: V \rightarrow V: \alpha_1 + \cdots + \alpha_k \mapsto \alpha_j$$

where $\alpha_i \in W_i$. This is well-defined by the properties of direct sum. □